Homework 4

Question

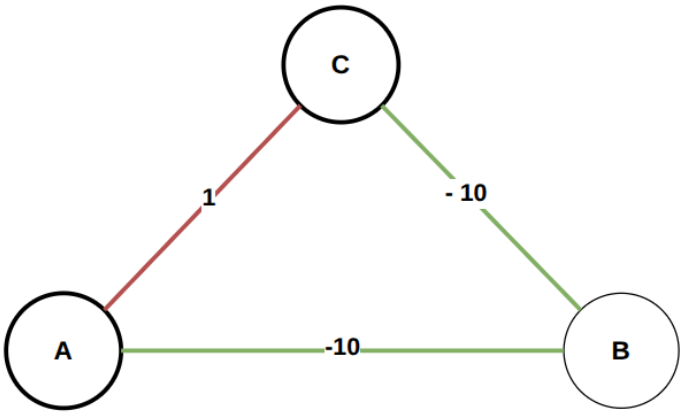
Assume that we have a graph with negative cost arcs

In order to reduce the graph to the case where all arcs have non negative costs we could sum to all arcs a suitably large constant

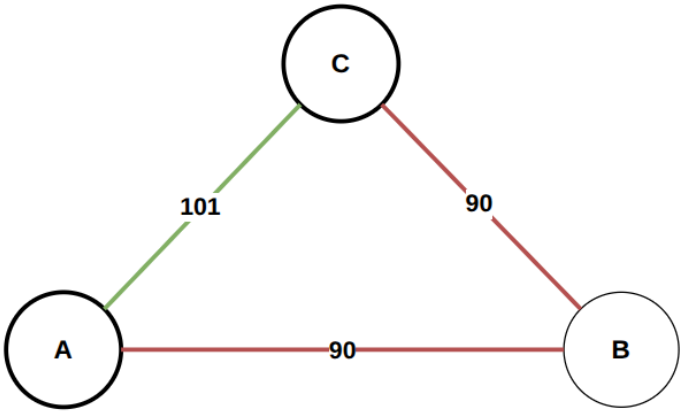
Provide a small instance where the shortest path between two nodes with the original costs
is different from the shortest path with the scaled costs
or prove that the two problems are equivalent

We can provide the following example:

Find the shortest path from node A to node C



The shortest path between node A and node C is the series of arcs AB, BC (total cost = -20), the only other option is the arc AC that has a more expensive cost (1).
If we were to change the graph by scaling everything by a suitably large constant k (for example, $k = 100$), we would obtain the following graph:



Now the shortest path between A and C is different: the path made by the arcs AB, BC costs 180, the path made by the arc AC costs 101. The new shortest path differs from the previous one, proving that scaled costs do not effectively maintain the properties of a graph.